

SEYONG KIM

CURRENT POSITION

Ph.D. Student in Electrical and Electronic Engineering
Yonsei University, Seoul, South Korea

Mar. 2023 – Present

CONTACT INFORMATION

Email: sykim@yonsei.ac.kr
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RESEARCH INTERESTS

Analysis and design of future wireless network systems, including advanced MIMO optimization frameworks, stochastic-geometry-based modeling of wireless networks, low-resolution ADC/DAC systems, and physical-layer design for virtualized RAN.

EDUCATION

Ph.D. in Electrical and Electronic Engineering
Yonsei University, Seoul, South Korea
Advisor: Prof. Jeonghun Park

Mar. 2023 – Present

M.S. in Electronic and Electrical Engineering
Kyungpook National University, Daegu, South Korea
Thesis: “Downlink NOMA for Short-Packet Internet-of-Things Communications with Low-Resolution ADCs”
Advisor: Prof. Jeonghun Park

Mar. 2021 – Feb. 2023

B.S. in Electronic and Electrical Engineering
Kyungpook National University, Daegu, South Korea
(Including mandatory military service in South Korea, Mar. 2015 – Mar. 2017)

Mar. 2014 – Feb. 2021

GRADUATE COURSES

Optimization Theory; Advanced Linear Algebra and Applications; Multiple Antennas Communication Systems; Broadband Wireless Communication Systems; Information Theory; Communication Network Analysis and Optimization; Low Earth Orbit Satellite Communication Systems; Management and Innovation of IT Technology.

WORKING EXPERIENCES

Visiting Scholar
The University of Texas at Austin, Austin, TX, USA
Host: Prof. Jeffrey G. Andrews

Mar. 2026 – Sep. 2026

Graduate Research Assistant
Yonsei University, Seoul, South Korea

Mar. 2024 – Present

- Design efficient space–time adaptive beamforming schemes for high-mobility satellite communications (funded by Samsung, Korea).
- Contribute to the development of 6G AI-native integrated RAN–core networks (funded by IITP, Korea).
- Develop a vRAN platform for aerial networks (funded by IITP, Korea).
- Participate in the 6G Cloud Research and Education Open Hub project (funded by IITP, Korea).

Teaching Assistant
Yonsei University, South Korea

- Design and Analysis of Advanced MIMO Systems *Fall 2024*
- Communication Systems *Fall 2023*
- Probability and Random Variables *Spring 2023*

Kyungpook National University, South Korea

- Statistical Inference and Machine Learning
- Communication Theory
- Information Theory
- Probability and Random Variables

Fall 2022

Fall 2021, Fall 2022

Spring 2021

Spring 2021

AWARDS & HONORS

Outstanding Graduate Student Award (Top Prize)

2025

Yonsei University, Seoul, South Korea

JOURNAL PUBLICATIONS

[3] S. Kim and J. Park, “Asymptotic Scaling Law Analysis of Multicast Satellite Communications with Massive MIMO,” *IEEE Wireless Communications Letters*, 2025.

[2] S. Kim, J. Choi, W. Shin, N. Lee, and J. Park, “Multibeam Satellite Communications with Massive MIMO: Asymptotic Performance Analysis and Design Insights,” *IEEE Transactions on Wireless Communications*, 2025.

[1] S. Kim, J. Choi, and J. Park, “Downlink NOMA for Short-Packet Internet-of-Things Communications with Low-Resolution ADCs,” *IEEE Internet of Things Journal*, 2023.